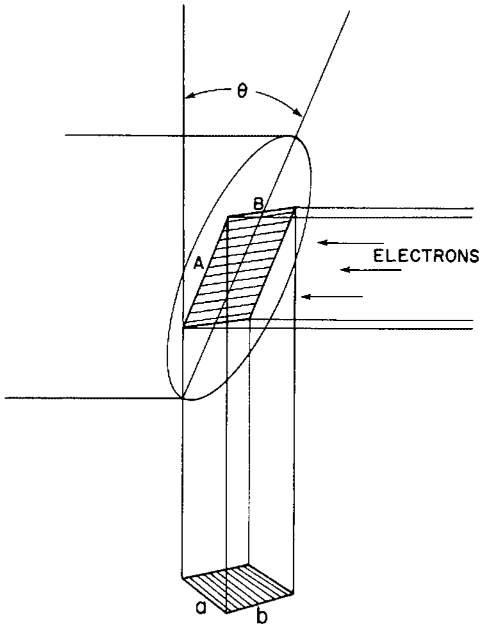
Electrons accelerated from the filament to the target are absorbed in the first 0.5-mm thickness of target. In this distance, a typical electron will experience 1000 or more interactions with target atoms.

the target of an x-ray tube is mounted at a steep angle with respect to the direction of impinging electrons (see Margin Figure 5-10). With the target at this angle, x rays appear to originate within a focal spot much smaller than the volume of the target absorbing energy from the impinging electrons. This apparent reduction in size of the focal spot is termed the *line-focus principle.* Most diagnostic x-ray tubes use a target angle between 6 and 17 degrees. In the illustration, side a of the projected or apparent focal spot may be calculated by

$$a = A \sin\theta \tag{5-4}$$

where A is the corresponding dimension of the true focal spot and θ is the target angle.



MARGIN FIGURE 5-10
Illustration of the line-focus principle, which reduces the apparent size of the focal spot.

Example 5-4

By using the illustration in the margin and Eq. (5-4), calculate a if $A = 7$ mm and $\theta = 17$ degrees.

$$\begin{aligned} a &= \sin\theta \\ &= (7\text{ mm})(\sin 17\text{ degrees}) \\ &= (7\text{ mm})(0.29) \\ &= 2\text{ mm} \end{aligned}$$

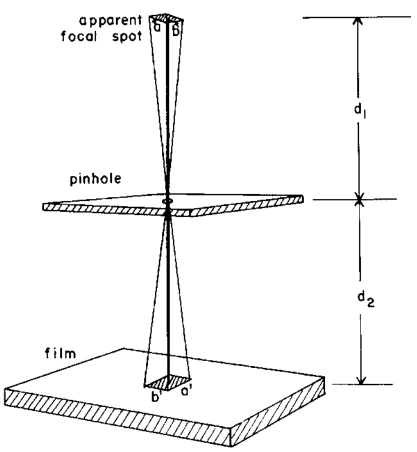
Side b of the apparent focal spot equals side B of the true focal spot because side B is perpendicular to the electron beam. However, side B is shorter than side A of the true focal spot because the width of a filament is always less than its length. When viewed in the center of the field of view, the apparent focal spot usually is approximately square.

As mentioned earlier, dual-focus diagnostic x-ray tubes furnish two apparent focal spots, one for fine-focus radiography (e.g., 0.6 mm^2 or less) produced with a smaller filament, and another for course-focus radiography (e.g., 1.5 mm^2) produced with a larger filament. The apparent focal spot to be used is determined by the tube current desired. The small filament is used when a low tube current (e.g., 100 mA) is satisfactory. The coarse filament is used when a larger tube current (e.g., 200 mA or greater) is required to reduce exposure time. Apparent focal spots of very small dimensions (e.g., $\leq$0.1 mm) are available with certain x-ray tubes used for magnification radiography.

The apparent size of the focal spot of an x-ray tube may be measured with a pinhole x-ray camera.[5,6] A hole with a diameter of a few hundredths of a millimeter is drilled in a plate that is opaque to x rays. The plate is positioned between the x-ray tube and film. The size of the image of the hole is measured on the exposed film. From the dimensions of the image and the position of the pinhole, the size of the apparent focal spot may be computed. For example, the dimension (a) of the apparent focal spot shown in the margin may be computed from the corresponding dimension (a') in the image by

$$a = a'\left(\frac{d_1}{d_2}\right) \tag{5-5}$$

where d_1 is the distance from the target to the pinhole and d_2 is the distance from the pinhole in the film.



MARGIN FIGURE 5-11
Pinhole method for determining the size *ab* of the apparent focal spot.

Focal spot size also can be measured with a resolution test object such as the star pattern shown in Figure 5-9. The x-ray image of the pattern on the right reveals a blur zone where the spokes of the test pattern are indistinct. From the diameter of the blur zone, the effective size of the focal spot can be computed in any dimension. This effective focal spot size may differ from pinhole camera measurements of the focal spot along the same dimension because the diameter of the blur zone is influenced not only by the actual focal spot dimensions, but also by the distribution of x-ray

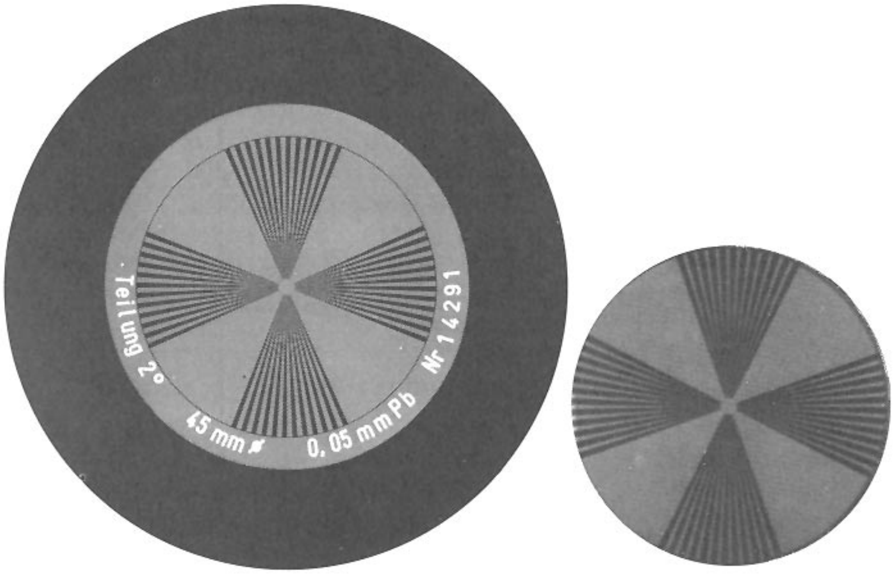
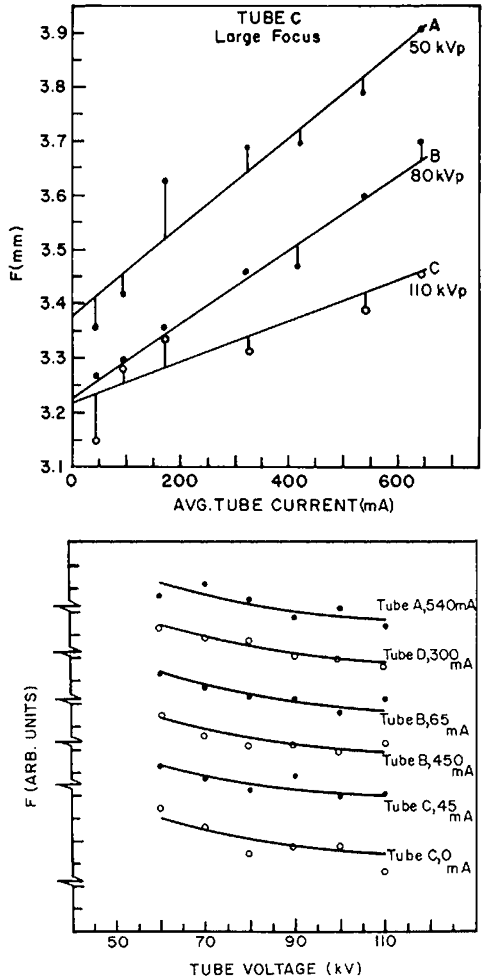


FIGURE 5-9
Contact radiograph (**left**) and x-ray image (**right**) of a star test pattern. The effective size of the focal spot may be computed from the diameter of the blur zone in the x-ray image.

intensity across the focal spot. In most diagnostic x-ray tubes, this distribution is not uniform. Instead, the intensity tends to be concentrated at the edges of the focal spot in a direction perpendicular to the electron beam.

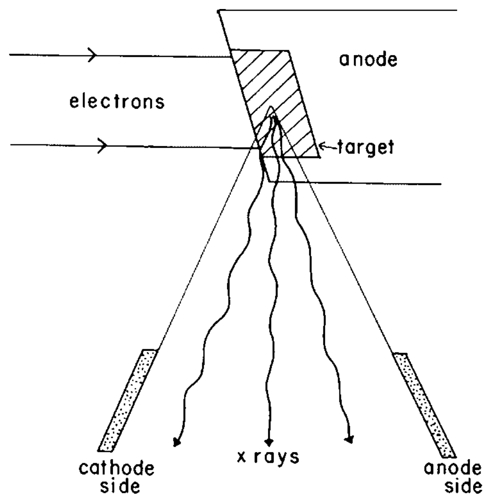
For most x ray tubes, the size of the focal spot is not constant. Instead, it varies with both the tube current and the voltage applied to the x-ray tube.[7,8] This influence is shown in the margin for the dimension of the focal spot parallel to the motion of impinging electrons. On the top, the growth or "blooming" of the focal spot with tube current is illustrated. The gradual reduction of the same focal spot dimension with increasing levels of peak voltage is shown on the bottom.



MARGIN FIGURE 5-12
Influence of tube current (**top**) and tube voltage (**bottom**) on the focal spot in a direction parallel to the motion of impinging electrons. (From Chaney, E., and Hendee, W. *Med Phys* 1974; **1**:131.)

Low-energy x rays generated in a tungsten target are attenuated severely during their escape from the target. For targets mounted at a small angle, the attenuation is greater for x rays emerging along the anode side of the x-ray beam than for those emerging along the side of the beam nearest the cathode. Consequently, the x-ray intensity decreases from the cathode to the anode side of the beam. This variation in intensity across an x-ray beam is termed the *heel effect*. The heel effect is noticeable for x-ray beams used in diagnostic radiology, particularly for x-ray beams generated at low kVp, because the x-ray energy is relatively low and the target angles are steep. To compensate for the heel effect, a filter may be installed in the tube housing near the exit portal of the x-ray beam. The thickness of such a filter increases from the anode to the cathode side of the x-ray beam. Positioning thicker portions of a patient near the cathode side of the x-ray beam also helps to compensate for the heel effect.

The heel effect increases with the steepness of the target angle. This increase limits the maximum useful field size obtainable with a particular target angle. For example, a target angle no steeper than 12 degrees is recommended for x-ray examinations using 14- ×17-in. film at a 40-in. distance from the x-ray tube, whereas targets as steep as 7 degrees may be used if field sizes no larger than 10 × 10 in. are required at the same distance.



MARGIN FIGURE 5-13
The heel effect is produced by increased attenuation of x rays in the sloping target near the anode side of the x-ray beam.

SPECIAL-PURPOSE X-RAY TUBES

Many x-ray tubes have been designed for special applications. A few of these special-purpose tubes are discussed in this chapter.

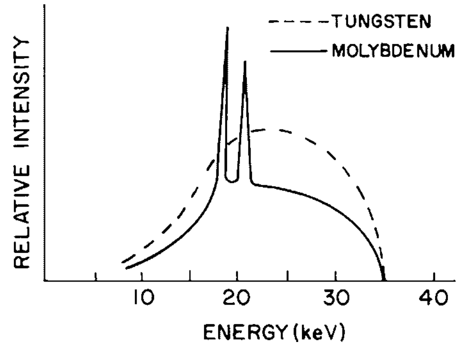


FIGURE 5-10
X-ray spectrum from molybdenum (solid line) and tungsten (dashed line) target x-ray tubes.

Grid-Controlled X-Ray Tubes

Grid-controlled x-ray tubes are also referred to as grid-pulsed or grid-biased x-ray tubes.

Grid-controlled x-ray tubes cost significantly more than non biased tubes.

In a grid-controlled x-ray tube, the focusing cup in the cathode assembly is maintained a few hundred volts negative with respect to the filament. In this condition, the negative potential of the focusing cap prevents the flow of electrons from filament to target. Only when the negative potential is removed can electrons flow across the x-ray tube. That is, applying and then momentarily removing the potential difference between the focusing cup and the filament provides an off-on switch for the production of x rays. Grid-controlled x-ray tubes are used for very short ("pulsed") exposures such as those required during digital radiography and angiography.

Field-Emission X-Ray Tubes

In the field-emission x-ray tube, the cathode is a metal needle with a tip about 1 μm in diameter. Electrons are extracted from the cathode by an intense electrical field rather than by thermionic emission. At diagnostic tube voltages, the rate of electron extraction is too low to provide tube currents adequate for most examinations, and efforts to market field-emission x-ray tubes in clinical radiology have been limited primarily to two applications: (1) pediatric radiography where lower tube currents can be tolerated and (2) high-voltage chest radiography where higher tube voltages can be used (as much as 300 kVp) to enhance the extraction of electrons from the cathode. Neither application has received much acceptance in clinical radiology.

Molybdenum-Target X-Ray Tubes

For low-voltage studies of soft-tissue structures (e.g., mammography), x-ray tubes with molybdenum and rhodium targets are preferred over tubes with tungsten targets. In the voltage range of 25 to 45 kVp, K-characteristic x rays can be produced in molybdenum but not in tungsten. These characteristic molybdenum photons yield a concentration of x rays on the low-energy side of the x-ray spectrum (Figure 5-10), which enhances the visualization of soft-tissue structures. Properties of molybdenum-target x-ray tubes are discussed in greater detail in Chapter 7.

RATINGS FOR X-RAY TUBES

The high rate of energy deposition in the small volume of an x-ray target heats the target to a very high temperature. Hence a target should have high thermal conductivity to transfer heat rapidly to its surroundings. Because of the high energy deposition in the target, rotating anodes are used in almost all diagnostic x-ray tubes. A rotating anode increases the volume of target material that absorbs energy from impinging electrons, thereby reducing the temperature attained by any portion of the anode. The anode is attached to the rotor of a small induction motor by a stem that usually is molybdenum. Anodes are 3 to 5 inches in diameter and rotate at speeds up to 10,000 rpm. The induction motor is energized for about 1 second before high voltage is applied to the x-ray tube. This delay ensures that electrons do not strike the target before the anode reaches its maximum speed of rotation. Energy deposited in the rotating anode is radiated to the oil bath surrounding the glass envelope of the x-ray tube.

Larger diameters and higher rotational speeds are required for applications such as angiography and helical-scan computed tomography that deliver greater amounts of energy to the x-ray target.

The rotating anode, together within the stator and rotor of the induction motor, are known collectively as the *anode assembly*.

Maximum Tube Voltage, Filament Current, and Filament Voltage

The maximum voltage to be applied between filament and target is specified for every x-ray tube. This "voltage rating" depends on the characteristics of the applied voltage (e.g., single phase, three phase, or constant potential) and on the properties of the x-ray tube (e.g., distance between filament and target, shape of the cathode assembly and target, and shape of the glass envelope). Occasional transient surges in voltage may be tolerated by an x-ray tube, provided that they exceed the voltage rating by no more than a few percent.

Limits are placed on the current and voltage delivered to coarse and fine filaments of an x-ray tube. The current rating for the filament is significantly lower for continuous compared with pulsed operation of the x-ray tube, because the temperature of the filament rises steadily as current flows through it.

Maximum Energy

Maximum-energy ratings are provided for the target, anode, and housing of an x-ray tube.[10,11] These ratings are expressed in heat units, where for single-phase electrical power

$$\begin{aligned}\text{Number of heat units (HU)} &= \text{(Tube voltage) (Tube current) (Time)}\\ &= \text{(kVp) (mA) (sec)} \qquad \textbf{(5-6)}\end{aligned}$$

If the tube voltage and current are constant, then 1 HU = 1J of energy. For three-phase power, the number of heat units is computed as

$$\begin{aligned}\text{Number of heat units (HU)} &= \text{(Tube voltage) (Tube current) (Time) (1.35)}\\ &= \text{(kVp) (mA) (sec) (1.35)} \qquad \textbf{(5-7)}\end{aligned}$$

For x-ray tubes supplied with single-phase (1φ), full-wave rectified voltage, the peak current through the x-ray tube is about 1.4 times the average current. The average current nearly equals the peak current in x-ray generators supplied with three-phase (3φ) voltage. For this reason, the number of heat units for an exposure from a 3φ generator is computed with the factor 1.35 in Eq. (5-7). For long exposures or a series of exposures with an x-ray tube supplied with 3φ voltage, more energy is delivered to the target, and the number of exposures in a given interval of time must be reduced. Separate rating charts are usually provided for 1φ and 3φ operation of an x-ray tube.

X-ray tubes are also rated in terms of their maximum power loading in kilowatts. Representative maximum power loadings are shown below[9]:

Focal Spot (mm)	Power Rating (kW)
1.2–1.5	80–125
0.8–1.0	50–80
0.5–0.8	40–60
0.3–0.5	10–30
≤ 0.3	1–10
≤ 0.1	<1

Energy ratings for the anode and the tube housing are expressed in terms of heat storage capacities. The heat storage capacity of a tube component is the total number of heat units that may be absorbed without damage to the component. Anode heat storage capacities for diagnostic x-ray tubes range from several hundred thousand to over a million heat units.

The heat storage capacity of the x-ray tube housing is also important because heat is transferred from the anode to the tube housing. The housing heat storage capacity exceeds the anode capacity and is usually on the order of 1.5 million HU.

To determine whether the target of an x-ray tube might be damaged by a particular combination of tube voltage, tube current, and exposure time, *energy rating charts* furnished with the x-ray tube should be consulted. To use the sample chart shown in Figure 5-11, a horizontal line is drawn from the desired tube current on the y axis to the curve for the desired tube voltage. From the intersection of the line and the curve, a vertical line dropped to the x axis reveals the maximum exposure time

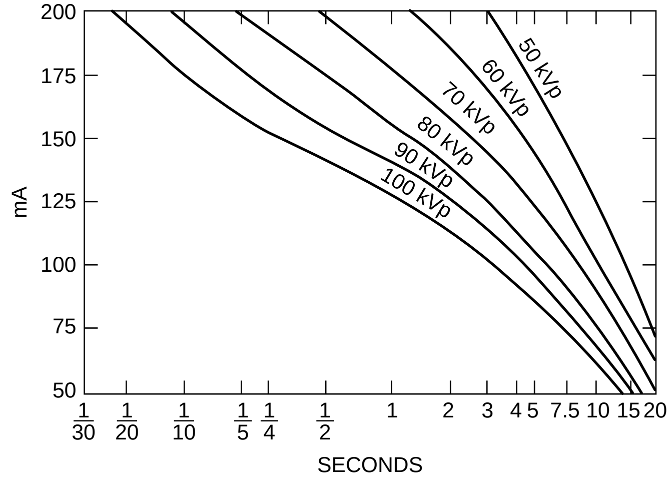


FIGURE 5-11
Energy rating chart for a Machlett Dynamax "25" x-ray tube with a 1-mm focal spot and single-phase, fully rectified voltage.(Courtesy of Machlett Laboratories, Inc.)

that can be used for a single exposure without possible damage to the x-ray target. The area under each voltage curve encompasses combinations of tube current and exposure time that do not exceed the target-loading capacity when the x-ray tube is operated at that voltage. The area above each curve reflects combinations of tube current and exposure time that overload the x-ray tube and might damage the target. Often switches are incorporated into an x-ray circuit to prevent the operator from exceeding the energy rating for the x-ray tube. Shown in Figure 5-12 are a few targets damaged by excess load or improper rotation of the target.

An *anode thermal-characteristics chart* describes the rate at which energy may be delivered to an anode without exceeding its capacity for storing heat (Figure 5-13). Also shown in the chart is the rate at which heat is radiated from the anode to the insulating oil and housing. For example, the delivery of 425 HU per second to the anode of the tube exceeds the anode heat storage capacity after 5.5 minutes. The delivery of 340 HU per second could be continued indefinitely. The cooling curve in Figure 5-13 shows the rate at which the anode cools after storing a certain amount of heat.

The housing-cooling chart in Figure 5-14 depicts the rate at which the tube housing cools after storing a certain amount of heat. The rate of cooling with and without forced circulation of air is shown. Charts similar to those in Figures 5-13 and 5-14 are used to ensure that multiple exposures in rapid succession do not damage an x-ray tube or its housing.

When several exposures are made in rapid succession, a target-heating problem is created that is not directly addressed in any of the charts described above. This target-heating problem is caused by heat deposition that exceeds the rate of heat dissipation in the focal track of the rotating anode. To prevent this buildup of heat from damaging the target, an additional tube-rating chart should be consulted. This chart, termed an angiographic rating chart because the problem of rapid successive exposures occurs frequently in angiography, is illustrated in Figure 5-15. Use of this chart is depicted in Example 5-9.

Example 5-5

From the energy-rating chart in Figure 5-11, is a radiographic technique of 150 mA, 1 second at 100 kVp permissible?

The maximum exposure time is slightly longer than 0.25 seconds for 150 mA at 100 kVp. Therefore, the proposed technique is unacceptable.

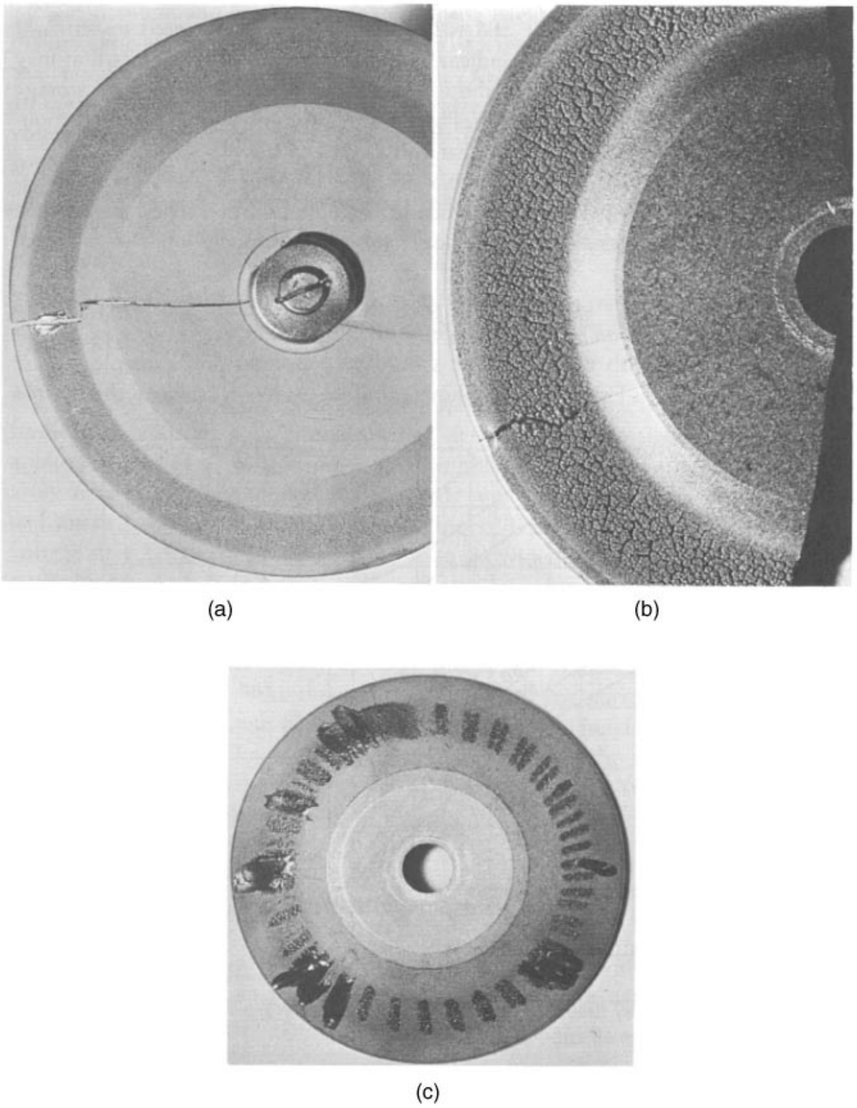


FIGURE 5-12
Rotating targets damaged by excessive loading or improper rotation of the target. **A:** Target cracked by lack of rotation. **B:** Target damaged by slow rotation and excessive loading. **C:** Target damaged by slow rotation.

Is 100 mA at 100 kVp for 1.5 seconds permissible?

The maximum exposure time is 3 seconds for 100 mA at 100 kVp. Therefore the proposed technique is acceptable.

Example 5-6

Five minutes of fluoroscopy at 4 mA and 100 kVp are to be combined with eight 0.5-second spot films at 100 kVp and 100 mA. Is the technique permissible according to Figures 5-11 and 5-13?

The technique is acceptable according to the energy-rating chart in Figure 5-11. By rearranging Eq. (5-6) we calculate that the rate of delivery of energy to the anode during fluoroscopy is (100 kVp) (4 mA) = 400 HU per second. From Figure 5-13, after 5 minutes approximately 60,000 HU have been accumulated by the anode. The eight spot films contribute an additional 40,000 HU [(100 kVp) (100 mA) (0.5 sec) 8 = 40,000 HU]. After all exposures have been made, the total heat stored in the

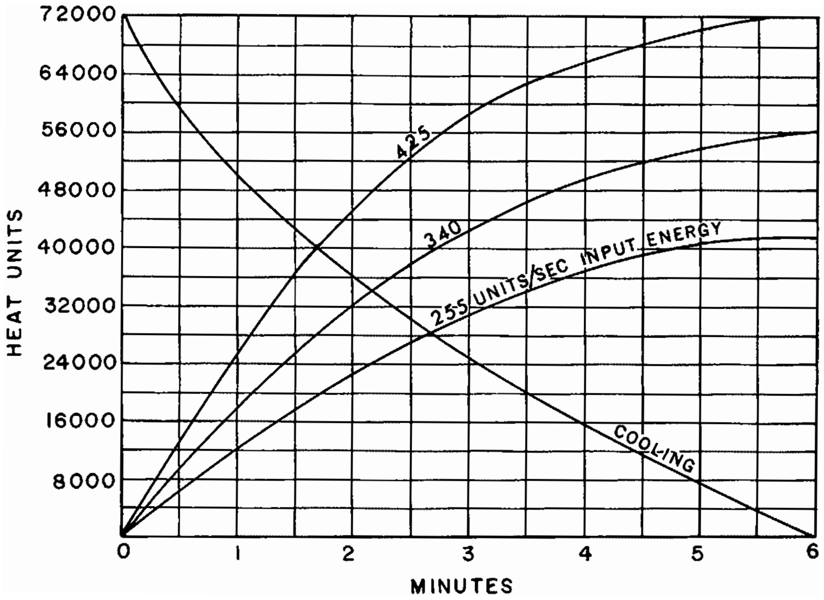


FIGURE 5-13
Anode thermal characteristics chart for a Machlett Dynamax "25" rotating anode x-ray tube. The anode heat-storage capacity is 72,000 HU. (Courtesy of Machlett Laboratories, Inc.)

anode is $40{,}000 + 60{,}000 = 100{,}000$ HU. This amount of heat exceeds the anode heat storage capacity of 72,000 HU. Consequently the proposed technique is unacceptable.

Example 5-7

Three minutes of fluoroscopy at 3 mA and 85 kVp are combined with four 0.25-second spot films at 85 kVp and 150 mA. From Figure 5-13, what time must elapse before the procedure may be repeated?

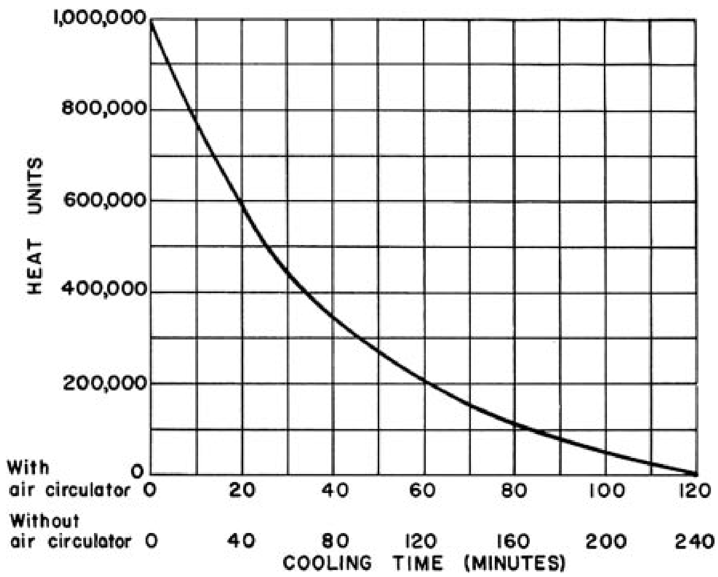


FIGURE 5-14
Housing-cooling chart for a Machlett Dynamax "25" x-ray tube. (Courtesy of Machlett Laboratories, Inc.)

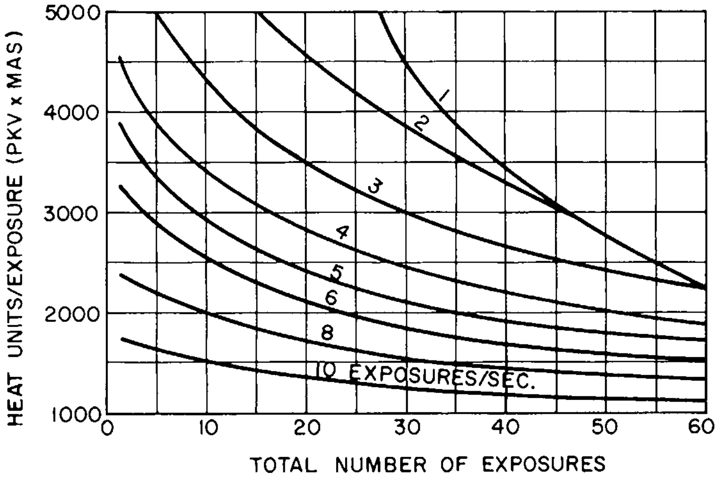


FIGURE 5-15
Angiographic rating chart for a Machlett Super Dynamax x-ray tube, 1.0-mm focal spot, full-wave rectification, single phase.

The rate of delivery of energy to the anode is (85 kVp) (3 mA) = 255 HU per second, resulting in a heat load of 31,000 HU after 3 minutes. To this heat load is added (85 kVp) (150 mA) (0.25 sec) (4) = 12,750 HU for the four spot films to yield a total heat load of 43,750 HU. From the position of the vertical axis corresponding to this heat load, a horizontal line is extended to intersect the anode-cooling curve at 1.4 minutes. From this intersection, the anode must cool until its residual heat load is 72,000 – 43,750 = 28,250 HU, so that when the 43,750 HU from the next procedure is added to the residual heat load, the total heat load does not exceed the 72,000 HU anode heat storage capacity. The time corresponding to a residual heat load of 28,250 HU is 2.6 minutes. Hence the cooling time required between procedures is 2.6 – 1.4 = 1.2 minutes.

Example 5-8

From Figures 5-11 and 5-13, it is apparent that three exposures per minute are acceptable if each 1φ exposure is taken at 0.5 second, 125 mA, and 100 kVp. Could this procedure be repeated each minute for 1 hour? The rate of energy transfer to the housing is

$$= (100 \text{ kVp}) (125 \text{ mA}) (0.5 \text{ sec}) (3 \text{ exposures/min})$$

$$= 18,750 \text{ HU/min}$$

At the end of 1 hour, (18,750 HU/min) (60 min/hr) = 1,125,000 HU will have been delivered to the housing. The heat storage capacity of the housing is only 1 million HU. Without forced circulation of air, the maximum rate of energy dissipation from the housing is estimated to be approximately 12,500 HU per minute (Figure 5-14). With air circulation, the rate of energy dissipation is 25,000 HU per minute. Therefore the procedure is unacceptable if the housing is not air-cooled, but is acceptable if the housing is cooled by forced circulation of air.

Example 5-9

From Figure 5-15, how many consecutive exposures can be made at a rate of six exposures per second if each exposure is taken at 85 kVp, 500 mA, and 0.05 seconds?

Each exposure produced (85 kVp) (500 mA) ().05 sec) = 2125 HU. A horizontal line from this position on the y axis intersects the 6 exposures per second curve at a position corresponding to 20 exposures. Hence, no more than 20 exposures should be made at a rate of 6 exposures per second.

PROBLEMS

5-1. Explain why tube current is more likely to be limited by space charge in an x-ray tube operated at low voltage and high tube current than in a tube operated at high voltage and low current. Would you expect the tube current to be space-charge limited or filament-emission limited in an x-ray tube used for mammography? What would you expect for an x-ray tube used for chest radiography at 120 kVp and 100 mA?

*5-2. How many electrons flow from the cathode to the anode each second in an x-ray tube with a tube current of 50 mA? (1A = 1 coulomb/sec.) If the tube voltage is constant and equals 100 kV, at what rate (joules per second) is energy delivered to the anode?

5-3. Explain why off-focus radiation is greatly reduced by shutters placed near the target of the x-ray tube.

*5-4. An apparent focal spot of 1 mm is projected from an x-ray tube. The true focal spot is 5 mm. What is the target angle? Why is the heel effect greater in an x-ray beam from a target with a small target angle?

*5-5. From Figure 5-11, is a radiographic technique of 125 mA for 3 seconds at 90 kVp permissible? Is a 4-second exposure at 90 kVp and 100 mA permissible? Why does the number of milliampere-seconds permitted for a particular voltage increase as the exposure time is increased?

*5-7. From Figure 5-14, how many exposures are permitted each minute over a period of 1 hour if each exposure is made for 1 second at 100 mA and 90 kVp? Is the answer different if the tube is operated with forced circulation of air?

*5-8. What kinetic energy do electrons possess when they reach the target of an x-ray tube operated at 250 kVp? What is the approximate ratio of bremsstrahlung to characteristic radiation produced by these electrons? Calculate the minimum wavelength of x-ray photons generated at 250 kVp.

*5-9. A lead plate is positioned 20 in. from the target of a diagnostic x-ray tube. The plate is 50 in. from a film cassette. The plate has a hole 0.1 mm in diameter. The image of the hole is 5 mm. What is the size of the apparent focal spot?

*5-10. The target slopes at an angle of 12 degrees in a diagnostic x-ray tube. Electrons are focused along a strip of the target 2 mm wide. How long is the strip if the apparent focal spot is 2×2 mm?

*For problems marked with an asterisk, answers are provided on p. 491.

SUMMARY

- Conventional x-ray tubes contain the following components:
  - Heated filament
  - Cathode assembly
  - X-ray target
  - High voltage supply
  - Rectification circuitry
  - Filtration
  - Glass envelope
- The apparent size of the x-ray focal spot depends on
  - Filament size
  - Tube current and voltage
  - Target angle
  - Position along the anode-cathode axis
- X-ray production efficiency is increased by
  - Voltage rectification
  - Use of three-phase and constant-potential tube voltage
  - Increased tube voltage
  - Higher-Z target
- Special-purpose x-ray tubes include
  - Grid-controlled
  - Field emission
  - Mammography
- X-ray tube rating charts include
  - Energy rating charts
  - Anode thermal characteristic charts
  - Housing cooling charts
  - Angiographic rating charts

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CHAPTER

6

RADIATION QUANTITY AND QUALITY

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Medical Imaging Physics, Fourth Edition, by William R. Hendee and E. Russell Ritenour
ISBN: 0-471-38226-4

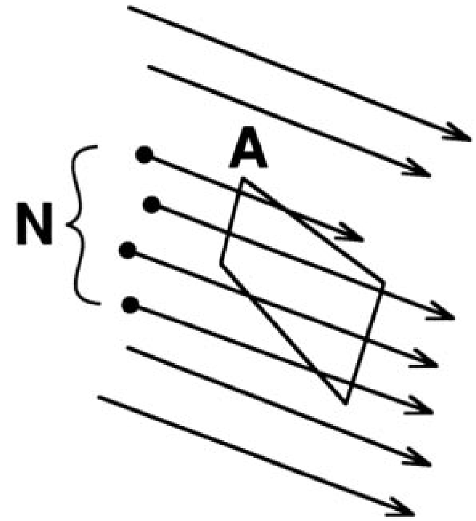
OBJECTIVES

After completing this chapter, the reader should be able to:

- Define the following terms: intensity, fluence, flux.
- Define and give the SI and traditional units for exposure, dose, RBE dose, dose equivalent, equivalent dose, and effective dose.
- Calculate the photon fluence required to obtain a unit exposure from the photon and energy fluence.
- Name seven methods that are used to measure radiation dose and describe how each method works.
- Explain the concept of electron equilibrium as it applies to ionization chambers.
- Calculate the effective atomic number of a composite material.
- Define the term "half value layer."
- Explain what is meant by the energy spectrum of an x-ray beam.

The specification and measurement of radiation demands assessment of the spatial and temporal distribution of the radiation. For some types of radiation the distribution of energies must also be considered. A complete description of the effects of radiation on a material requires analysis of how the various components of the radiation field interact with atoms and molecules of the material.

To avoid the burden of maintaining a large multidimensional database of detailed information, techniques have been developed that allow multiple interaction parameters to be summarized into singular values. These singular values—the rem, the sievert, the gray, and so on—are the radiation units described in this chapter. While they may be susceptible to the pitfalls of oversimplification, they are all attempts to provide a succinct answer to the question, What will happen when radiation strikes a material?



MARGIN FIGURE 6-1
The particle fluence of a beam of radiation is defined as the number of particles (N) passing through a unit area (A) that is perpendicular to the direction of the beam. If the beam is uniform, the location and size of the area A are arbitrary.

INTENSITY

The term *radiation* is defined as energy in transit from one location to another. The term *radiation intensity* is used colloquially to refer to a number of attributes of the output of a radiation source. In physics and engineering, the term "intensity" is defined more specifically in terms of energy per unit area, per unit time. In this text, the term rdiation intensity is given a specific definition.

A beam of radiation is depicted in Margin Figure 6-1. The radiation *fluence* of the beam Φ is defined as the number N of particles or photons per area A:

$$\Phi = \frac{N}{A} \tag{6-1}$$

If the beam is uniform, then the location or size of the area A is irrelevant so long as it is in the beam and perpendicular to the direction of the beam. If the beam is not uniform over its entire area, then the fluence must be averaged over a number of small areas, or specified separately for each area. The time rate of change of fluence, known as the radiation flux ϕ, is

$$\phi = \frac{\Phi}{t} = \frac{N}{A \cdot t} \tag{6-2}$$

If the fluence varies with time, then the flux must be averaged over time or specified at some instant.

If all particles or photons in the radiation beam possess the same energy, the energy fluence Ψ is simply the product of the radiation fluence Φ and the energy E per particle or photon:

$$\Psi = \Phi E = \frac{NE}{A} \tag{6-3}$$

Radiation flux is sometimes termed the *radiation flux density*.

TABLE 6-1 Fluence and Flux (Intensity) of a Beam of Radiation[a]

Quantity	*Symbol*	*Definition[b]*	*Units*
Particle (photon) fluence	Φ	$\frac{N}{A}$	$\frac{\text{Particles (photons)}}{\text{m}^2}$
Particle (photon) flux	ϕ	$\frac{N}{At}$	$\frac{\text{Particles (photons)}}{\text{m}^2 \cdot \text{sec}}$
Energy fluence	Ψ	$\frac{NE}{A}$	$\frac{\text{MeV}}{\text{m}^2}$
Energy flux (intensity)	ψ	$\frac{NE}{At}$	$\frac{\text{MeV}}{\text{m}^2 \cdot \text{sec}}$

[a] *Note:* These expressions assume that the number of particles or photons does not vary over time or over the area A, and that all particles or photons have the same energy.

[b] N = number of particles or photons; E = energy per particle or photon; A = area; t = time.

Similarly, the particle or photon flux ϕ may be converted to the energy flux ψ, also known as the intensity I, by multiplying by the energy E per particle or photon:

$$I = \psi = \phi E = \frac{NE}{At} \tag{6-4}$$

If the radiation beam consists of particles or photons having different energies $(E_1, E_2, \ldots, E_m)$, then the intensity (or energy flux) is determined by

$$I = \psi = \sum_{i=1}^{m} f_i \phi E_i \tag{6-5}$$

where f_i represents the fraction of particles having energy E_i, and the symbol

$$\sum_{i=1}^{m}$$

indicates that the intensity is determined by adding the components of the beam at each of m energies. The concepts of radiation fluence and flux are summarized in Table 6-1.

Expressions similar to Eqs. (6-1) to (6-5) may be derived for any type of particle beam such as α, β, neutron, or high-energy nuclei. In the case of electromagnetic radiation such as x or γ rays, the photons have an energy $E = h\nu$. In the case of radiation that is described in terms of waves (such as ultrasound or radio waves), the definition of intensity (Equation 6-4) is modified as

$$I = \frac{E_t}{At} \tag{6-6}$$

where E_t is the total energy delivered by the wave during the time t.

Example 6-1

An abdominal radiograph uses 10^{13} photons to expose a film with an area of 0.15 m^2 (1.5×10^{-1} m^2 or 1500 cm^2) during an exposure time of 0.1 second. All photons have an energy of 40 keV. Find the photon fluence Φ, the photon flux ϕ, the energy fluence Ψ, and the intensity I.

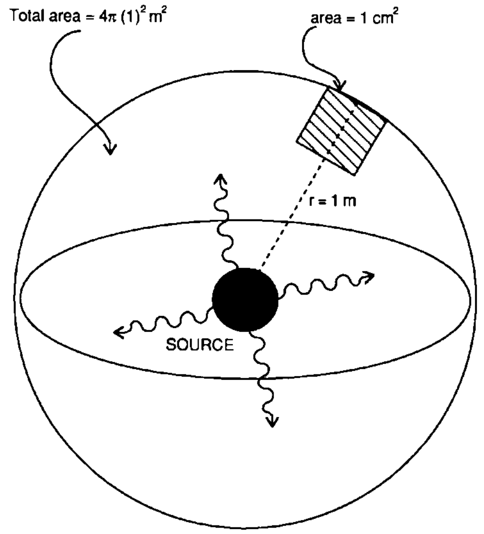
$$\text{Photon fluence } \Phi = \frac{N}{A} = \frac{1 \times 10^{13} \text{ photons}}{1.5 \times 10^{-1} \text{ m}^2} = 6.7 \times 10^{13} \ \frac{\text{photons}}{\text{m}^2}$$

$$\text{Photon flux } \phi = \frac{N}{At} = \frac{\Phi}{t} = \frac{6.7 \times 10^{13} \ \frac{\text{photons}}{\text{m}^2}}{0.1 \ \text{sec}} = 6.7 \times 10^{14} \ \frac{\text{photons}}{\text{m}^2 \cdot \text{sec}}$$

$$\text{Energy fluence } \Psi = \frac{NE}{A} = \Phi E = \left(6.7 \times 10^{13}\ \frac{\text{photons}}{\text{m}^2}\right)\left(40 \frac{\text{keV}}{\text{photon}}\right)$$
$$= 2.68 \times 10^{15}\ \frac{\text{keV}}{\text{m}^2}$$
$$= 2.68 \times 10^{12}\ \frac{\text{MeV}}{\text{m}^2}$$

$$\text{Intensity } I = \Psi = \frac{NE}{At} = \frac{\Psi}{t} = \frac{2.68 \times 10^{12}\ \frac{\text{MeV}}{\text{m}^2}}{0.1\ \text{sec}} = 2.68 \times 10^{13}\ \frac{\text{MeV}}{\text{m}^2 \cdot \text{sec}}$$

Note: A beam produced by an x-ray tube actually contains a spectrum of energies (Chapter 5), and a more accurate estimate of intensity would involve a weighted sum of the contributions of the various photon energies to the total intensity. The next example shows a source of radiation that produces two different photon energies and also demonstrates the concept of detector geometry—that is, the effect of area.



MARGIN FIGURE 6-2
Photons emitted in any direction from an isotopic source eventually cross the surface of a sphere.

Example 6-2

A radionuclide releases 270-keV photons in 90% of its decays and 360-keV photons in approximately 10% of its decays. When 10^6 photons have been released by decay of the radionuclide, what is the photon fluence and energy fluence over a 1-cm^2 area at a distance of 1 m from the source?

Because photons are emitted isotropically (i.e., with equal probability in any direction) from the source, 10^6 photons cross the surface of a sphere with a radius of 1 m (Margin Figure 6-2). The number of photons that cross an area of 1 cm^2 (10^{-4} m^2) on the sphere surface is the fraction of the 10^6 photons that are intercepted by the 1-cm^2 area. The surface area of a sphere is $4\pi r^2$, where r is the sphere's radius.

$$\text{Fraction of total emissions} = \frac{10^{-4}\ \text{m}^2}{4\pi(1\ \text{m})^2} = 7.96 \times 10^{-6}$$

$$\text{Photons crossing the 1-cm}^2\text{ area} = (7.96 \times 10^{-6})10^6 = 7.96\ \text{ photons}$$

That is, the photon fluence is approximately 8 photons/cm^2.

The energy fluence must be weighted according to the fraction of photons having each of the two possible energies.

$$\Psi = \sum_{i=1}^{2} f_i \Phi E_i$$
$$= (0.9)\left(8\ \frac{\text{photons}}{\text{cm}^2}\right)(270\ \text{keV}) + (0.1)\left(8\ \frac{\text{photons}}{\text{cm}^2}\right)(360\ \text{keV})$$
$$= (1944 + 288)\ \frac{\text{keV}}{\text{cm}^2}$$
$$= 2.23\ \frac{\text{MeV}}{\text{cm}^2}$$

Although photon and energy flux densities and fluences are important in many computations in radiologic physics, these quantities cannot be measured easily. Hence, units have been defined that are related more closely to common methods for measuring radiation quantity.

TRADITIONAL VERSUS SYSTÈME INTERNATIONAL UNITS

The measurement of x and γ radiation has presented technical challenges from the time of their discovery. In his initial studies, Röntgen used the blackening of a photographic emulsion (i.e., film), as a "dosimeter."[1] Nonlinearity and energy dependence of the film's response caused difficulties. The "erythema dose" (the amount

TABLE 6-2 Traditional (T) and Système International (SI) Quantities

Quantity	*Unit* *T*	*SI*	*To Convert from T to SI, Multiply by*
Exposure	Roentgen (R)	Coulomb/kg (C/kg)	2.58×10^{-4}
Absorbed dose	rad	Gray (Gy)	0.01
Absorbed dose equivalent	rem	Sievert (Sv)	0.01

of radiation required to produce reddening of the skin) was for many years the major dosimetric method for evaluating the effects of radiation treatment. Biologic variability and the lack of an objective measure of "reddening" were major problems with this technique.

In 1928, the roentgen (R) was defined as the principal unit of radiation quantity for x rays of medium energy. The definition of the roentgen has been revised many times, with each revision reflecting an increased understanding of the interactions of radiation as well as improvements in the equipment used to detect these interactions. Over this same period, several other units of radiation quantity have been proposed. In 1958, the International Commission on Radiation Units and Measurements (ICRU) organized a continuing study of the units of radiation quantity. Results of this continuing study are described in various ICRU reports.[2]

The units described in earlier ICRU reports are considered the "traditional" system (T) of units. Just as the United States is converting to the "metric" system of units, radiology is converting to the system of radiation units known as the Système International (SI).[3] Both the traditional and SI systems are used in this text with preference given to SI units. Table 6-2 gives a summary of units and conversion factors between them.

RADIATION EXPOSURE

Primary ion pairs (electrons and positive ions) are produced as ionizing radiation interacts with atoms of an attenuating medium. Secondary ion pairs are produced as the primary ion pairs dissipate their energy by ionizing nearby atoms. The total number of ion pairs produced is proportional to the energy that the radiation deposits in the medium. The concept of *radiation exposure* is based on the assumption that the absorbing medium is air. If Q is the total charge (negative or positive) liberated as x or γ rays interact in a small volume of air of mass m, then the radiation exposure X at the location of the small volume is

$$X = \frac{Q}{m} \tag{6-7}$$

The total charge reflects the production of both primary and secondary ion pairs, with the secondary ion pairs produced both inside and outside of the small volume of air. The traditional unit of radiation exposure is the *roentgen* (R):

$$1\text{R} = 2.58 \times 10^{-4} \text{ coulomb/kg air}$$

This definition of the roentgen is numerically equivalent to an older definition:

$$\begin{aligned} 1\text{R} &= 1 \text{ electrostatic unit (ESU)}/0.001293 \text{ g air} \\ &= 1 \text{ ESU/cm}^3 \text{ air at STP*} \end{aligned}$$

*STP = standard temperature (0°C) and pressure (1 atm or 760 mm Hg).

The roentgen is applicable only to x and γ radiation and to photon energies less than about 3 MeV. That is, the roentgen cannot be used for particle beams or for beams of high-energy photons. For photon energies above 3 MeV, it becomes increasingly difficult to determine how many secondary ion pairs are produced outside of the measurement volume as a result of interactions occurring in the volume, and vice versa. There is no specially named unit for radiation exposure in SI. Instead, the fundamental units of coulomb and kilogram are used (see Table 6-2). The unit roentgen and the concept of radiation exposure in general are disappearing from use as more fundamental expressions of radiation quantity gain acceptance.

Example 6-3

Find the energy absorbed in air from an exposure of 1 coulomb/kg. The W-quantity, the average energy required to produce ion pairs in a material (Chapter 4), is 33.85 eV per ion pair (IP) for air. The energy absorbed during an exposure X of 1 coulomb per kilogram is

$$\begin{aligned} E_x &= \frac{(1 \text{ coulomb/kg})(33.85 \text{ eV/IP})(1.6 \times 10^{-19} \text{ J/eV})}{(1.6 \times 10^{-19} \text{ coulomb/IP})} \\ &= 33.85 \text{ J/kg} \end{aligned} \tag{6-8}$$

Thus, for every coulomb per kilogram of exposure, air absorbs 33.85 J/kg of energy.

Energy and Photon Fluence per Unit Exposure

From the definitions of energy and photon fluence and some fundamental quantities that have been determined for air, we may calculate the energy or photon fluence required to produce a given amount of exposure. From Eq. (6-8), the energy absorbed in air is $33.85X$ (J/kg) during an exposure to X coulomb/kg. The absorbed energy can also be stated as

$$\begin{aligned} \text{Energy absorbed in air} &= (\text{Energy fluence})\ (\text{Total mass energy absorption coefficient}) \\ &= \Psi\left(\frac{\text{J}}{\text{m}^2}\right)(\mu_{\text{en}})_m\left(\frac{\text{m}^2}{\text{kg}}\right) \\ &= \Psi(\mu_{\text{en}})_m\left(\frac{\text{J}}{\text{kg}}\right) \end{aligned} \tag{6-9}$$

where $(\mu_{\text{en}})_m$ is the total mass energy absorption coefficient for x- or γ-ray photons that contribute to the energy fluence. The coefficient $(\mu_{\text{en}})_m$ is defined as (Chapter 4)

$$(\mu_{\text{en}})_m = \mu_m\left(\frac{E_a}{h\nu}\right),$$

where μ_m is the total mass attenuation coefficient of air for photons of energy $h\nu$ and E_a represents the average energy transformed into kinetic energy of electrons and positive ions per photon absorbed or scattered from the x- or γ-ray beam.[4] The average energy E_a is corrected for characteristic x rays radiated from the attenuating medium, as well as for bremsstrahlung produced as electrons interact with nuclei within the attenuating medium. Mass energy absorption coefficients for a few media, including air, are listed in Table 6-3.

By combining Eqs. (6-8) and (6-9), we obtain

$$\Psi(\mu_{\text{en}})_m = 33.85\ X$$

TABLE 6-3 Mass Energy Absorption Coefficients for Selected Materials and Photon Energies

	Mass Energy Absorption Coefficient $(\mu_{en})_m$ (m^2/kg)			
Photon Energy (MeV)	*Air*	*Water*	*Compact Bone*	*Muscle*
0.01	0.466	0.489	1.90	0.496
0.02	0.0516	0.0523	0.251	0.0544
0.03	0.0147	0.0147	0.0743	0.0154
0.04	0.00640	0.00647	0.0305	0.00677
0.05	0.00384	0.00394	0.0158	0.00409
0.06	0.00292	0.00304	0.00979	0.00312
0.08	0.00236	0.00253	0.00520	0.00255
0.10	0.00231	0.00252	0.00386	0.00252
0.20	0.00268	0.00300	0.00302	0.00297
0.30	0.00288	0.00320	0.00311	0.00317
0.40	0.00296	0.00329	0.00316	0.00325
0.50	0.00297	0.00330	0.00316	0.00327
0.60	0.00296	0.00329	0.00315	0.00326
0.80	0.00289	0.00321	0.00306	0.00318
1.0	0.00280	0.00311	0.00297	0.00308
2.0	0.00234	0.00260	0.00248	0.00257
3.0	0.00205	0.00227	0.00219	0.00225
4.0	0.00186	0.00205	0.00199	0.00203
5.0	0.00173	0.00190	0.00186	0.00188
6.0	0.00163	0.00180	0.00178	0.00178
8.0	0.00150	0.00165	0.00165	0.00163
10.0	0.00144	0.00155	0.00159	0.00154

Source: National Bureau of Standards Handbook 85. Washington, D.C., U.S. Government Printing Office, 1964.

The energy fluence per unit exposure (Ψ/X) is

$$\frac{\Psi}{X} = \frac{33.85}{(\mu_{\text{en}})_m} \tag{6-10}$$

where $(\mu_{\text{en}})_m$ is expressed in units of m^2/kg, Ψ in J/m^2, and X in coulombs/kg.

For monoenergetic photons, the photon fluence per unit exposure, Φ/X, is the quotient of the energy fluence per roentgen divided by the energy per photon:

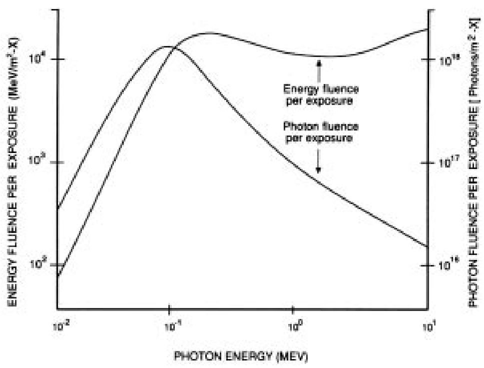
$$\frac{\Phi}{X} = \frac{\Psi}{h\nu\left(1.6 \times 10^{-13}\ \text{J/MeV}\right)}$$

From Eq. (6-10), we obtain

$$\frac{\Phi}{X} = \frac{2.11 \times 10^{14}}{h\nu\,(\mu_{\text{en}})_m} \tag{6-11}$$

with $h\nu$ expressed in MeV and Φ in units of photons/m^2.

The photon and energy fluence per unit exposure are plotted in Margin Figure 6-3 as a function of photon energy. At lower photon energies, the large influence of photon energy upon the energy absorption coefficient of air is reflected in the rapid change in the energy and photon fluence per unit exposure. Above 100 keV, the energy absorption coefficient is relatively constant, and the energy fluence per unit exposure does not vary greatly.[5] However, the photon fluence per unit exposure decreases steadily as the energy per photon increases.



MARGIN FIGURE 6-3
Photon and energy fluence per exposure, plotted as a function of the photon energy in MeV. Exposure is expressed in coulomb/kg. To convert to the traditional unit, roentgen, multiply the vertical scales by 2.58×10^{-4}.

Example 6-4

Compute the energy and photon fluence per unit exposure for ^{60}Co γ rays. The average energy of the photons is 1.25 MeV, and the total energy absorption coefficient is 2.67×10^{-3} m^2/kg (Table 6-3).

$$\frac{\Psi}{X} = \frac{33.85}{(\mu_{\text{en}})_m}$$
$$= \frac{33.85}{2.67 \times 10^{-3}\ \text{m}^2/\text{kg}}$$
$$= 12{,}600\ \frac{\text{J}}{\text{m}^2} \qquad \textbf{(6-10)}$$

$$\frac{\Phi}{X} = \frac{2.11 \times 10^{14}}{h\nu\,(\mu_{\text{en}})_m}$$
$$= \frac{2.11 \times 10^{14}}{(1.25\ \text{MeV/photon})(2.67 \times 10^{-3}\ \text{m}^2/\text{kg})}$$
$$= 6.32 \times 10^{16}\ \frac{\text{photons}}{\text{m}^2} \qquad \textbf{(6-11)}$$

UNITS OF RADIATION DOSE

The Gray

The traditional unit of radiation dose, the rad, is an acronym for "radiation absorbed dose."

Chemical and biological changes in tissue exposed to ionizing radiation depend upon the energy absorbed in the tissue from the radiation, rather than upon the amount of ionization that the radiation produces in air. To describe the energy absorbed in a medium from any type of ionizing radiation, the quantity of radiation should be described in SI units of gray (or traditional units of rads). The gray (Gy) is a unit of *absorbed dose** and represents the absorption of one joule of energy per kilogram of absorbing material:

$$1\ \text{Gy} = 1\ \text{J/kg}$$

The absorbed dose D in gray delivered to a small mass m in kilograms is

$$D(\text{Gy}) = \frac{E/m}{1\ \text{J/kg}} \qquad \textbf{(6-12)}$$

The traditional unit of radiation dose, the rad, represents an absorption of 100 ergs of energy per gram of absorbing material.

where E, the absorbed energy in joules, is "the difference between the sum of the energies of all the directly and indirectly ionizing particles which have entered the volume, and the sum of the energies of all those which have left it, minus the energy equivalent of any increase in rest mass that took place in nuclear or elementary particle reactions within the volume." [6]

This definition means that E is the total energy deposited in a small volume of irradiated medium, corrected for energy removed from the volume in any way.

Example 6-5

The name of the quantity "kerma" is an acronym for the phrase "kinetic energy released in a material" with an "a" added for convenience.

If a dose of 0.05 Gy (5 centigray [cGy]) is delivered uniformly to the uterus during a diagnostic x-ray examination, how much energy is absorbed by each gram of the uterus?

$$D = \frac{E/m}{1\ \text{J/kg-Gy}}$$

*The Gy is also a unit of kerma, defined as the sum of the initial kinetic energies of all charged particles liberated by ionizing radiation in a volume element, divided by the mass of matter in the volume element. Under conditions of charged-particle equilibrium with negligible energy loss by bremsstrahlung, kerma and absorbed dose are identical. The output of x-ray tubes is sometimes described in terms of "air kerma" expressed in terms of energy deposited per unit mass of air.

$$
\begin{aligned}
E &= (1 \text{ J/kg-Gy})(D)(m) \\
&= (1 \text{ J/kg-Gy})(0.05 \text{ Gy})\left(10^{-3}\ \frac{\text{kg}}{\text{g}}\right)(1\ g) \\
&= 5 \times 10^{-5} \text{ J}
\end{aligned}
$$

During an exposure of 1 C/kg, the energy absorbed per kilogram of air is

$$1 \text{ C/kg} = 33.85 \text{ J/kg in air}$$

or, in traditional units,

$$
\begin{aligned}
1 \text{ R} &= 2.58 \times 10^{-4} \text{ C/kg} \\
&= (2.58 \times 10^{-4} \text{ C/kg})\left(33.85\ \frac{\text{J/kg}}{\text{C/kg}}\right) \\
&= 86.9 \times 10^{-4} \text{ J/kg in air}
\end{aligned}
$$

Because 1 rad $= 10^{-2}$ J/kg, we have

$$
\begin{aligned}
1 \text{ C/kg} &= 3385 \text{ rad in air} \\
1 \text{ R} &= (2.58 \times 10^{-4} \text{ C/kg})(3385 \text{ rad-kg/C}) \\
&= 0.869 \text{ rad in air}
\end{aligned}
$$

The Sievert

At the present time, there are five different quantities that use the SI unit of sievert and the traditional unit of rem. In many situations, they are approximately equal in magnitude. The differences lie in the ways that one or more of the four variables that influence the extent of biological damage are managed. These four variables reflect the:

- Damage at the cellular level per unit dose by different types of radiation (alpha, beta, x rays, neutrons, etc.)
- Sensitivity of different body tissues
- Effects that damage to different tissues have upon the overall health of the organism
- Impact upon future generations

RBE Dose

Chemical and biological effects of irradiation depend not only upon the amount of energy absorbed in an irradiated medium, but also upon the distribution of the absorbed energy within the medium. For equal absorbed doses, various types of ionizing radiation often differ in the efficiency with which they elicit a particular chemical or biological response. The *relative biologic effectiveness* (RBE) describes the effectiveness or efficiency with which a particular type of radiation evokes a certain chemical or biological effect. The relative biological effectiveness is computed by comparing results obtained with the radiation in question with those obtained with a reference radiation (e.g., medium-energy x rays or ^{60}Co radiation):

$$\text{RBE} = \frac{\text{Dose of reference radiation required to produce a particular response}}{\text{Dose of radiation in question required to produce the same response}} \tag{6-13}$$

TABLE 6-4 RBE[a] of ^{60}Co Gammas, with Different Biologic Effects Used as a Criterion for Measurement[b]

Effect	*RBE of ^{60}Co Gammas*	*Source*
30-day lethality and testicular atrophy in mice	0.77	Storer et al. (7)
Splenic and thymic atrophy in mice	1	Storer et al. (7)
Inhibition of growth in *Vicia faba*	0.84	Hall (8)
LD_{50}[a] in mice, rat, chick embryo, and yeast	0.82–093	Sinclair (9)
Hatchability of chicken eggs	0.81	Loken et al. (10)
HeLa cell survival	0.90	Krohmer (11)
Lens opacity in mice	0.8	Upton et al. (12)
Cataract induction in rats	1	Focht et al. (13)
L-cell survival	0.76	Till and Cunningham (14)

[a] RBE, relative biological effectiveness; $LD_{50,}$ median lethal dose.

[b] The RBE of 200-kV x rays is taken as 1.

For a particular type of radiation, the RBE may vary from one chemical or biological response to another. Listed in Table 6-4 are the results of investigations of the relative biological effectiveness of ^{60}Co radiation. For these data, the reference radiation is medium-energy x rays. It is apparent that the RBE for ^{60}Co γ rays varies from one biological response to the next.

The RBE dose in sievert (Sv) is the product of the RBE and the dose in gray:

$$\text{RBE dose (Sv)} = \text{Absorbed dose (Gy)} \times \text{RBE} \qquad \textbf{(6-14)}$$

The ICRU has suggested that the concept of RBE dose should be limited to descriptions of radiation dose in radiation biology.[6] That is, RBE dose pertains only to an exact set of experimental conditions, and it is not applicable to general radiation protection situations.

DOSE EQUIVALENT

Often the effectiveness with which different types of radiation produce a particular chemical or biological effect varies with the linear energy transfer (LET) of the radiation. The *dose-equivalent* (DE) in sievert is the product of the dose in Gy and a *quality factor* (QF), which varies with the LET of the radiation.

$$\text{DE (Sv)} = D(\text{Gy}) \times \text{QF} \qquad \textbf{(6-15)}$$

The dose equivalent reflects a recognition of differences in the effectiveness of different radiations to inflict overall biological damage and is used during computations associated with radiation protection.[15] Quality factors are listed in Table 6-5 as a function of LET, and in Table 6-6 for different types of radiation.

These quality factors should be used to determine shielding requirements and to compute radiation doses to personnel working with or near sources of ionizing radiation.

Example 6-6

A person receives an average whole-body dose of 1 mGy from ^{60}Co γ rays and 0.5 mGy from neutrons with an energy of 10 MeV. What is the dose equivalent to the

TABLE 6-5 Relation Between Specific Ionization, Linear Energy Transfer, and Quality Factor

Average Specific Ionization (IP/μm) in Water	*Average LET (keV/μm) in Water*	*QF*
100 or less	3.5 or less	1
100–200	3.5–7.0	1–2
200–650	7.0–23	2–5
650–1500	23–53	5–10
1500–5000	53–175	10–20

Source: Recommendations of the International Commission on Radiological Protection. *Br J Radiol* 1955; Suppl. 6.

person in millisieverts?

$$\begin{aligned} \text{DE (mSV)} &= [D(\text{mGy}) \times \text{QF}]_{\text{gammas}} + [D(\text{mGy}) \times \text{QF}]_{\text{neutrons}} \\ &= (1\text{ mGy})(1) + (0.5\text{ mGy})(20) \\ &= 11\text{mSv} \end{aligned}$$

Example 6-7

A person accidentally ingests a small amount of tritium (β-particle $E_{\max} = 0.018$ MeV). The average dose to the gastrointestinal tract is estimated to be 5 mGy. What is the dose equivalent in millisieverts?

From Table 6-6, the quality factor is 1.0 from tritium β-particles.

$$\begin{aligned} \text{DE (mSv)} &= D(\text{mGy}) \times \text{QF} \\ &= (5\text{ mGy})(1.0) \\ &= 5\text{ mSv} \end{aligned}$$

Effective Dose Equivalent

A further refinement of the prediction of bioeffects can be obtained by multiplying the dose equivalent by a factor to convert the dose to a point within a specific patient tissue to a measure of overall detriment to the patient. In the late 1970s, the available epidemiological data[16] were used to compile weighting factors for many tissues and organs that, when multiplied by the dose equivalent, would yield the effective dose equivalent. The effective dose equivalent, an estimate of overall detriment to the individual, was designed to account for cancer mortality in individuals between the ages of 20 and 60, as well as hereditary effects in the next two generations.[17] These weighting factors have been superseded by factors reformulated in the 1990s.

Equivalent Dose

In the early 1990s, up-to-date dosimetry and epidemiological results from studies of the survivors of the atomic bombs dropped on Hiroshima and Nagasaki in 1945 began to yield statistically robust analyses of the relative radiation sensitivities of different tissues, and the consequences of irradiation of these tissues to the organism as a whole.[18,19] This work also affected the assignment of factors to account for the energy and type of radiation. When the average dose to an organ is known, the risk can be assigned in terms of more recent estimates of fatality from cancer. In addition, the new factors consider the detriment from non fatal cancer and hereditary effects for future generations. In view of these changes, and to differentiate from the older system of dosimetry, the term equivalent dose was adopted to refer to the product

TABLE 6-6 Quality Factors for Different Radiations[a]

Type of Radiation	*QF*
X rays, γ rays, and β particles	1
Thermal neutrons	5
Neutrons and protons	20
α Particles from natural radionuclides	20
Heavy recoil nuclei	20

[a] These data should be used only for purposes of radiation protection. Data from National Council on Radiation Protection and Measurements. *NCRP Report*, no. 91, 1987.

Five quantities are expressed in units of sievert at the present time. It is expected that some of them will become obsolete at some time in the future. Listed below are the five quantities and the factors that are taken into account in specifying the value of those quantities in units of sievert.

1. RBE dose (relative biological effectiveness) - Relative effectiveness of different **types** of radiation in eliciting a **specific biologic effect**, such as reddening of the skin in **specific parts of the body** of animals or humane exposed in experimental conditions. Used in journal articles and textbooks.
2. Dose Equivalent - Relative effectiveness of different **types** of radiation in producing overall bioeffects using data from the 1970's. Still used in federal regulations.
3. Effective Dose Equivalent - Differing biological harm caused by different **types** of radiation and also the **region of the body** irradiated using data from the 1970's. Dose is defined at a point in the patient within each organ or organ system. Still used in federal regulations.
4. Equivalent Dose - Differing biological harm caused by different **types** of radiation using data from the 1990's. Used in current ICRP and NCRP recommendations.
5. Effective Dose - Differing biological harm caused by different **types** of radiation and also the **region of the body** irradiated using data from the 1990's. Used in current ICRP and NCRP recommendations.

Note: There is consistency in that each time that the term "equivalent" is used, the type of radiation is taken into account. Each time the word "effective" is used, both the type of radiation and the region of the body that is irradiated are taken into account.

TABLE 6-7 Radiation Weighting Factors

Source	W_R
X rays, gamma rays, electrons, positrons, muons at all energies	1
Neutrons	
<10 keV	5
10 keV to 100 keV	10
>100 keV to 2 MeV	20
>2 MeV to 20 MeV	10
>20 MeV	5
Protons, >2 MeV	2
Alpha particles, fission fragments, nonrelativistic heavy nuclei	20

Source: National Council on Radiation Protection and Measurements. Limitation of Exposure to Ionizing Radiation, Report No. 116. Bethesda, MD, NCRP, 1993.

of the average organ dose and the newly defined radiation weighting factors, W_R (Table 6-7).

Effective Dose

The newly defined tissue weighting factors, W_T, are given in Table 6-8. They are formulated to yield the effective dose by multiplying by an equivalent dose that represents an average organ dose that has been calculated using the newer radiation weighting factors W_R given in Table 6-7.

Example 6-8

As in Example 6-7, a person ingests a small amount of tritium. The average dose to the gastrointestinal tract is estimated to be 5 mGy. For the purposes of this example, we will assume that a negligible amount appears in the bladder. What is the equivalent dose in millisieverts? And what is the effective dose in millisieverts?

From Table 6-7, the radiation weighting factor is 1 for beta particles.

The equivalent dose is then

$$\begin{aligned} \text{ED (mSv)} &= D(\text{mGy}) \times W_R \\ &= (5\ \text{mGy})(1.0) \\ &= 5\ \text{mSv} \end{aligned}$$

Note that in this situation, 5 mSv is the same numerical value that was calculated for the dose equivalent in Example 6-7. This is because the radiation weighting factor (W_R) for beta particles equals the quality factor (QF) for beta particles. Also, a uniform dose deposition is assumed in this problem so that all point doses are assumed to equal the average organ doses. If the detailed dosimetry were known with greater precision, the values of point doses within organs might have led to a different calculation of dose equivalent.

From Table 6-7, the tissue weighting factor is 0.05 for the esophagus, 0.12 for the stomach, and 0.12 for the colon.

The effective dose is then

$$\begin{aligned} \text{ED (mSv)} &= D(\text{mGy}) \times W_R \times W_T \\ &= (5\ \text{mGy})(0.05) + (5\ \text{mGy})(0.12) + (5\ \text{mGy})(\ 0.12) \\ &= 0.25\ \text{mSv} + 0.60\ \text{mSv} + 0.60\ \text{mSv} \\ &= 1.45\ \text{mSv} \end{aligned}$$

Note that the effective dose is a smaller number than the equivalent dose, because only partial body irradiation has taken place. The value of 1.45 mSv is the dose that, if received uniformly over the whole body, would lead to the same detrimental results (probability of fatal cancers, nonfatal cancers, and hereditary disorders) as the dose actually received by the gastrointestinal tract.

TABLE 6-8 Tissue Weighting Factors

Tissue	W_T
Gonads	0.20
Active bone marrow	0.12
Colon	0.12
Lungs	0.12
Stomach	0.12
Bladder	0.05
Breasts	0.05
Esophagus	0.05
Liver	0.05
Thyroid	0.05
Bone surfaces	0.01
Skin	0.01
Remainder	0.05

Source: Recommendations of the International Commission on Radiological Protection, ICRP Publication 60, Pergamon Press, Elmsford, NY, 1991.

MEASUREMENT OF RADIATION DOSE

The absorbed dose to a medium describes the energy absorbed during exposure of the medium to ionizing radiation. A radiation dosimeter provides a measurable response to the energy absorbed from incident radiation. To be most useful, the dosimeter should absorb an amount of energy equal to what would be absorbed in the medium that the dosimeter displaces. For example, a dosimeter used to measure radiation dose in soft tissue should absorb an amount of energy equal to that absorbed by the same mass of soft tissue. When this requirement is satisfied, the dosimeter is said to be *tissue equivalent*. Few dosimeters are exactly tissue equivalent, and corrections must be applied to most direct measurements of absorbed dose in tissue.

Calorimetric Dosimetry

Almost all the energy absorbed from radiation is eventually degraded to heat. If an absorbing medium is insulated from its environment, then the rise in temperature of the medium is proportional to the energy absorbed. The temperature rise T may be measured with a thermocouple or thermistor.

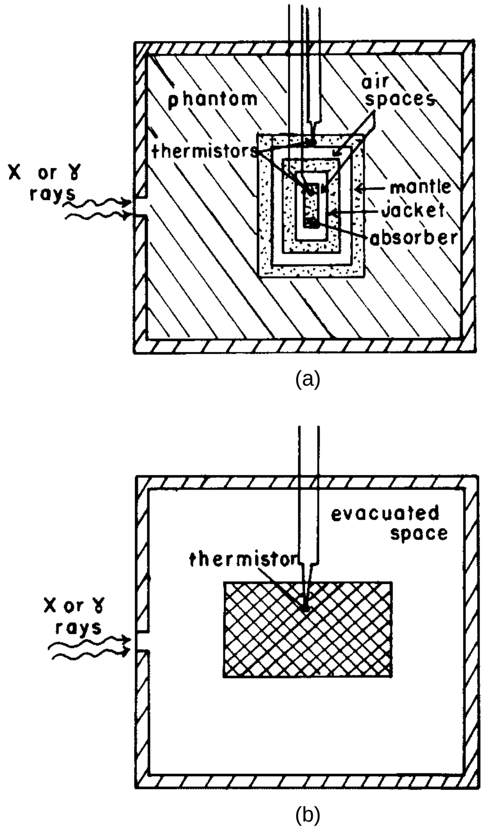
A thermocouple is a junction of dissimilar metals that yields a voltage that varies with temperature. A thermistor is a solid-state device with an electric resistance that changes rapidly with temperature.

A radiation calorimeter is an instrument used to measure radiation absorbed dose by detecting the increase in temperature of a mass of absorbing material.[20] The absorbed dose in the material is

$$D(\text{Gy}) = \frac{E}{m(1\ \text{J/kg-Gy})} = \frac{4.186\ (\text{J/calorie})s\,\Delta T}{(1\ \text{J/kg-Gy})} \qquad \textbf{(6-16)}$$

where E is the energy absorbed in joules, m is the mass of the absorber in kilograms, s is the specific heat of the absorber in cal/kg °C, and ΔT is the temperature rise in Celsius (centigrade) degrees. For Eq. (6-16) to be correct, the absorbing medium must be insulated from its environment. To measure the absorbed dose in a particular medium, the absorber of the calorimeter must possess properties similar to those of the medium. Graphite is often used as the absorbing medium in calorimeters designed to measure the absorbed dose in soft tissue. A calorimeter used to measure radiation absorbed dose is illustrated in Margin Figure 6-4.

If the absorbing medium is thick and dense enough to absorb nearly all of the incident radiation, then the increase in temperature reflects the total energy delivered to the absorber by the radiation beam. Calorimeters that measure the total energy in a beam of radiation usually contain a massive lead absorber.



MARGIN FIGURE 6-4
Radiation calorimeters: **A:** For the measurement of absorbed dose in soft tissue. **B:** For the measurement of the total energy in a beam of radiation.

Photographic Dosimetry

The emulsion of a photographic film contains crystals of a silver halide embedded in a gelatin matrix. When the emulsion is developed, metallic silver is deposited in regions that were exposed to radiation. Unaffected crystals of silver halide are removed during fixation. The amount of silver deposited at any location depends upon many factors, including the amount of energy absorbed by the emulsion at that location. The transmission of light through a small region of film varies inversely with the amount of deposited silver. The transmission T is

$$T = \frac{I}{I_0}$$

where I represents the intensity of light transmitted through a small region of film and I_0 represents the light intensity with the film removed. The transmission may be measured with an optical densitometer. Curves relating transmission to radiation exposure, dose, or energy fluence are obtained by measuring the transmission through films receiving known exposures, doses, or fluences. If the problems described below are not insurmountable, the radiation exposure to a region of blackened film may be determined by measuring the transmission of light though the region and referring to a calibration curve.

Accurate dosimetry is difficult with photographic emulsions for reasons including those listed below.[21]

1. The optical density of a film depends not only on the radiation exposure to the emulsion but also on variables such as the energy of the radiation and the conditions under which the film is processed. For example, the optical density of a photographic film exposed to 40-keV photons may be as great as the density of a second film receiving an exposure 30 times greater from photons of several MeV. Photographic emulsions are said to be "energy

dependent" because their response to x and γ rays of various energies differs significantly from the response of air and soft tissue. In most cases, calibration films used to relate optical density to radiation exposure must be exposed to radiation identical to that for which dosimetric measurements are needed. This requirement often is difficult to satisfy, particularly when films are exposed in phantoms or in other situations where a large amount of scattered radiation is present.

2. When compared with air or soft tissue, the high-Z emulsion of a photographic film attenuates radiation rapidly. Errors caused by differences in attenuation may be particularly severe when radiation is directed parallel to the plane of a single film or perpendicular to a stack of films.
3. Differences in the thickness and composition of the photographic emulsion may cause the radiation sensitivity to vary from one film to another.

Chemical Dosimetry

Oxidation or reduction reactions are initiated when certain chemical solutions are exposed to ionizing radiation. The number of molecules affected depends on the energy absorbed by the solution. Consequently, the extent of oxidation or reduction is a reflection of the radiation dose to the solution. These changes are the basis of chemical dosimetry.

The chemical dosimeter used most widely is ferrous sulfate. For photons that interact primarily by Compton scattering, the ratio of the energy absorbed in ferrous sulfate to that absorbed in soft tissue equals 1.024, the ratio of the electron densities (electrons per gram) of the two media. For photons of higher and lower energy, the ratio of energy absorption increases above 1.024 because more energy is absorbed in the higher-Z ferrous sulfate.

The use of solutions of ferrous sulfate to measure radiation dose was described by Fricke and Morse in 1929.[22] A solution of ferrous sulfate used for radiation dosimetry is sometimes referred to as a *Fricke dosimeter*.[23] Although this dosimeter is reliable and accurate (±3%), it is relatively insensitive and absorbed doses of 50 to 500 Gy are required before the oxidation of Fe^{2+} to Fe^{3+} is measurable. The Fricke dosimeter was recommended in earlier years for calibration of high-energy electron beams used in radiation therapy.[24]

The yield of a chemical dosimeter such as ferrous sulfate is expressed by the G value:

$$G = \text{Number of molecules affected per 100 eV absorbed}$$

The G value for the oxidation of Fe^{2+} to Fe^{3+} in the Fricke dosimeter varies with the LET of the radiation. Many investigators have assumed that G is about 15.4 molecules per 100 eV for a solution of ferrous sulfate exposed to high-energy electrons or x and γ rays of medium energy. After exposure to radiation, the amount of Fe^{3+} ion in a ferrous sulfate solution is determined by measuring the transmission of ultraviolet light (305 nm) through the solution. Once the number of affected molecules is known, the energy absorbed by the ferrous sulfate solution may be computed by dividing the number of molecules by the G value.

Example 6-9

A solution of ferrous sulfate is exposed to ^{60}Co γ radiation. Measurement of the transmission of ultraviolet light (305 nm) through the irradiated solution indicates the presence of Fe^{3+} ion at a concentration of 0.0001 mol/L. What was the radiation dose absorbed by the solution?

For a concentration of 0.0001 gram-molecular weight/L and with a density of 10^{-3} kg/mL for the ferrous sulfate solution, the number of Fe^{3+} ions per kilogram is

$$\begin{aligned}
&\text{Number of Fe}^{3+}\text{ ions/kg} \\
&\quad = \frac{(0.0001 \text{ gram-mol wt}/1000 \text{ mL})\,(6.02 \times 10^{23} \text{ molecules/gram-mol wt})}{10^{-3} \text{ kg/mL}} \\
&\quad = 6.02 \times 10^{19} \, \frac{\text{molecules}}{\text{kg}}
\end{aligned}$$

For a G value of 15.4 molecules per 100 eV for ferrous sulfate, the absorbed dose in the solution is

$$D(\text{rad}) = \frac{(6.02 \times 10^{19}\ \text{molecules/kg})(1.6 \times 10^{-19}\ \text{J/eV})}{(15.4\ \text{molecules/100 eV})(1\ \text{J/kg-Gy})}$$
$$= 63\ \text{Gy}$$

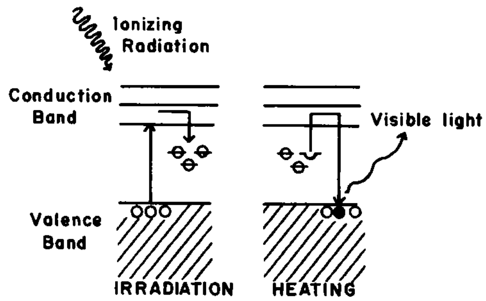
Scintillation Dosimetry

Certain materials fluoresce or "scintillate" when exposed to ionizing radiation. The rate at which scintillations occur depends upon the rate of absorption of radiation in the scintillator. With a solid scintillation detector (e.g., thallium-activated sodium iodide), a light guide couples the scintillator optically to a photomultiplier tube. In the photomultiplier tube, light pulses from the detector are converted into electronic signals.

Scintillation detectors furnish a measurable response at very low dose rates and respond linearly over a wide range of dose rates. However, most scintillation detectors contain high-Z atoms, and low-energy photons are absorbed more rapidly in the detectors than in soft tissue or air. This *energy dependence* is a major disadvantage of using scintillation detectors to measure radiation exposure or dose to soft tissue. A few organic scintillation detectors have been constructed that are air equivalent or tissue equivalent over a wide range of photon energies.

Thermoluminescence Dosimetry

Diagramed in Margin Figure 6-5 are energy levels for electrons within crystals of a thermoluminescent material such as LiF. Electrons "jump" from the valence band to the conduction band by absorbing energy from ionizing radiation impinging upon the crystals. Some of the electrons return immediately to the valence band; other are "trapped" in intermediate energy levels supplied by impurities in the crystals. The number of electrons trapped in intermediate levels is proportional to the energy absorbed by the LiF phosphor during irradiation. Only rarely do electrons escape from the traps and return directly to the ground state. Unless energy is supplied for their release, almost all of the trapped electrons remain in the intermediate energy levels for months or years after irradiation. If the crystals are heated, energy is supplied to release the trapped electrons. Released electrons return to the conduction band where they fall to the valance band. Light is released as the electrons return to the valence band. This light is directed onto the photocathode of a photomultiplier tube. Because the amount of light striking the photocathode is proportional to the energy absorbed in the LiF during irradiation, the signal from the photomultiplier tube increases with the radiation dose absorbed in the phosphor.

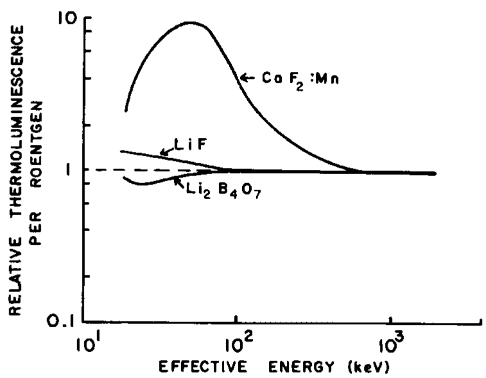


MARGIN FIGURE 6-5
Electron transitions occurring when thermoluminescent LiF is irradiated and heated.

The effective atomic number* of dosimetric LiF (8.18) is close to that for soft tissue (7.4) and for air (7.65).[25] Hence, for identical exposures to radiation, the amount of energy absorbed by LiF is reasonably close to that absorbed by an equal mass of soft tissue or air. Small differences between the energy absorption in LiF and air are reflected in the "energy dependence" curve for LiF shown in Margin Figure 6-6.

Lithium fluoride is used widely for the measurement of radiation dose within patients and phantoms, for personnel dosimetry, and for many other dosimetric measurements. Dosimetric LiF may be purchased as loose crystals, a solid extruded rod, solid chips, pressed pellets, or crystals embedded in a Teflon matrix.

Thermoluminescent dosimeters composed of CaF_2:Mn are frequently used for personnel monitoring and, occasionally, for other measurements of radiation dose. When compared with LiF and $Li_2B_4O_7$, CaF_2:Mn phosphor is more sensitive to ionizing radiation; however, the response of this phosphor varies rapidly with photon energy (Margin Figure 6-6).[26]



MARGIN FIGURE 6-6
Relative thermoluminescence from various materials per unit exposure is plotted as a function of the effective energy of incident x- or γ-ray photons. Data have been normalized to the response per unit exposure for ^{60}Co γ rays.

*The atomic number of a hypothetical element that attenuates photons at the same rate as the material in question.

Other Solid-State Dosimeters

Photoluminescent dosimeters are similar to thermoluminescent dosimeters except that ultraviolet light rather than heat causes the dosimeters to emit light. Most photoluminescent dosimeters are composed of silver-activated metaphosphate glass of either high-Z or low-Z composition. The response from both types of glass increases rapidly as the energy of incident x or γ rays is reduced. Shields have been designed to reduce the energy dependence of these materials.[27] For exposure rates up to at least 10^8 R/sec, the photoluminescent response of silver-activated metaphosphate glass is independent of exposure rate. Photoluminescent dosimeters are occasionally used for personnel monitoring and other dosimetric measurements.

Radiation-induced changes in the optical density of glasses and plastics have been used occasionally to measure radiation quantity. Changes in the optical density are determined by measuring the transmission of visible or ultraviolet light through the material before and after exposure to radiation. Silver-activated phosphate glass, cobalt-activated borosilicate glass, and transparent plastics have been used as radiation dosimeters.[28] Glass and plastic dosimeters are very insensitive to radiation, and high exposures are required to obtain a measurable response.

To predict the interactions of the vast number of particles produced during nuclear fission, Stanislaw Ulam and John Von Neumann developed a mathematical formalism in 1946 in which the outcomes of individual interactions were determined at random according to the probabilities involved. Large numbers of these "events" were computed, and the results were summed by what was then a relatively new device, the electronic computer. The results were used to predict the overall behavior of the radiations involved in nuclear fission and, later, nuclear fusion. Because of the association of this technique with random outcomes, the developers named it the "Monte Carlo" method, after the famous gaming casino in Monaco.

Ionization Measurements

With designation of the energy absorbed per kilogram of air as E_a, Eq. (6-9) may be rewritten as

$$\Psi = \frac{E_a}{(\mu_{\text{en}})_m}$$

In a unit volume of air, the energy absorbed per kilogram is

$$E_a = \frac{E}{\rho}$$

where E is the energy deposited in the unit volume by impinging x or γ radiation and ρ is the density of air (1.29 kg/m^3 at STP). If J is the number of primary and secondary ion pairs (IPs) produced as a result of this energy deposition, then

$$E = J\,W$$

where $W = 33.85$ eV/IP. The energy fluence Ψ is

$$\Psi = \frac{J\,W}{\rho\,(\mu_{\text{en}})_m} \tag{6-17}$$

Example 6-10

A 1-m^3 volume of air at STP is exposed to a photon fluence of 10^{15} photons per square meter. Each photon possesses 0.1 MeV. The total energy absorption coefficient of air is 2.31×10^{-3} m^2/kg for 0.1-MeV photons. How many ion pairs (IPs) are produced inside and outside of the 1-cm^3 volume? How much charge is measured if all the ion pairs are collected?

$$\begin{aligned}\Psi &= \Phi E && (6\text{-}3')\\ &= \left(10^{15}\,\frac{\text{photons}}{\text{m}^2}\right)\left(0.1\,\frac{\text{MeV}}{\text{photon}}\right)\left(1.6\times 10^{-13}\,\frac{\text{J}}{\text{MeV}}\right)\\ &= 1.6\times 10^{1}\,\frac{\text{J}}{\text{m}^2}\\ &= \frac{J\,W}{\rho\,(\mu_{\text{en}})_m}\end{aligned}$$

$$J = \frac{\Psi\rho\,(\mu_{\text{en}})_m}{W} \tag{6-18}$$

$$= \frac{(1.6\times10^1\ \text{J/m}^2)(1.29\ \text{kg/m}^3)(2.31\times10^{-3}\ \text{m}^2/\text{kg})(10^{-6}\ \text{m}^3/\text{cm}^3)(1\ \text{cm}^3)}{(33.8\ \text{eV/IP})(1.6\times10^{-19}\ \text{J/eV})}$$

$$= 88\times10^8\ \text{ion pairs}$$

The charge Q collected is

$$Q = (88\times10^8\ \text{IP})\left(1.6\times10^{-19}\ \frac{\text{coulomb}}{\text{IP}}\right)$$

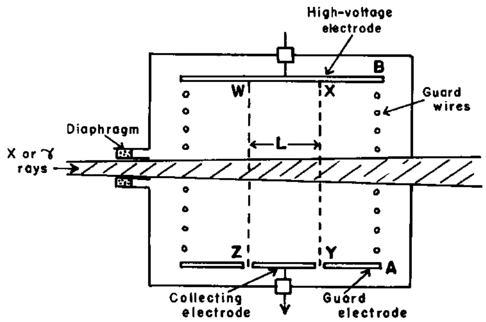
$$= 1.41\times10^{-9}\ \text{coulomb}$$

It is not possible to measure all of the ion pairs resulting from the deposition of energy in a small volume of air exposed to x or γ radiation. In particular, secondary ion pairs may escape measurement if they are produced outside the "collecting volume" of air. However, a small volume may be chosen so that energy lost outside the collecting volume by ion pairs created within equals energy deposited inside the collecting volume by ion pairs that originate outside. When this condition of *electron equilibrium* is satisfied, the number of ion pairs collected inside the small volume of air equals the total ionization J. The principle of electron equilibrium is used in the free-air ionization chamber.

Free-Air Ionization Chamber

Free-air ionization chambers are used by standards laboratories to provide a fundamental measurement of ionization in air. These chambers are often used as a reference in the calibration of simpler dosimeters such as thimble chambers described later in this chapter.

X or γ rays incident upon a free-air ionization chamber are collimated by a tungsten or gold diaphragm into a beam with cross-sectional area A (Margin Figure 6-7). Inside the chamber, the beam traverses an electric field between parallel electrodes A and B, with the potential of electrode B highly negative. Electrode A is grounded electrically and is divided into two guard electrodes and a central collecting electrode. The guard electrodes ensure that the electric field is uniform over the collecting volume of air between the electrodes. To measure the total ionization accurately, the range of electrons liberated by the incident radiation must be less than the distance between each electrode and the air volume exposed directly to the x- or γ-ray beam. Furthermore, for electron equilibrium to exist, the photon flux density must remain constant across the chamber, and the distance from the diaphragm to the border of the collecting volume must exceed the electron range. If all of these requirements are satisfied, then the number of ion pairs liberated by the incident photons per unit volume of air is



MARGIN FIGURE 6-7
A free-air ionization chamber. The collecting volume of length L is enclosed within the region WXYZ. The air volume exposed directly to the x- or γ-ray beam is depicted by the hatched area.

$$\frac{\text{IP}}{\text{Unit volume}} = \frac{N}{AL}$$

where N is the number of ion pairs collected, A is the cross-sectional area of the beam at the center of the collecting volume, and L is the length of the collecting volume. The charge Q (positive or negative) collected by the chamber is

$$Q = N\left(1.6\times10^{-19}\ \frac{\text{coulomb}}{\text{IP}}\right)$$

Because 1 R equals 2.58×10^{-4} coulomb/kg air, the number of roentgens X corresponding to a charge Q in a free-air ionization chamber is

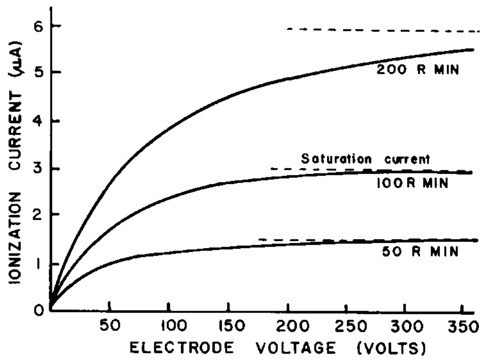
$$X = \left(\frac{1}{2.58\times10^{-4}\ \text{coulomb/kg-R}}\right)\frac{Q}{AL\rho} \tag{6-19}$$

where ρ is the density of air.

TABLE 6-9 Range and Percentage of Total Ionization Produced by Photoelectrons and Compton Electrons for X Rays Generated at 100, 200, and 1000 kVp

X-Ray Tube Voltage (kVp)	*Photoelectrons*		*Compton Electrons*		*Electrode Separation in Free Air Ionization Chamber*
	Range in Air (cm)	*Percentage of Total Ionization*	*Range in Air (cm)*	*Percentage of Total Ionization*	
100	12	10	0.5	90	12 cm
200	37	0.4	4.6	99.6	12 cm
1000	290	0	220	100	4 m

Source: Meredith, W., and Massey, J. *Fundamental Physics of Radiology*. Baltimore, Williams & Wilkins, 1968. Used with permission.



MARGIN FIGURE 6-8
Current in a free-air ionization chamber as a function of the potential difference across the electrodes of the chamber. Saturation currents are shown for different exposure rates. Data were obtained from a chamber with electrodes 1 cm apart. (From Johns, H., and Cunningham, J. *The Physics of Radiology*, 3rd edition. Springfield, IL, Charles C Thomas, 1969. Used with permission.)

To prevent ion pairs from recombining, the potential difference between the electrodes of a free-air ionization chamber must be great enough to attract all ion pairs to the electrodes. A voltage this great is referred to as a *saturation voltage*. Ionization currents in a free-air chamber subjected to different exposure rates are plotted in Margin Figure 6-8 as a function of the potential difference across the electrodes. For any particular voltage, there is an exposure rate above which significant recombination occurs. Unless the potential difference is increased, an exposure rate higher than this limiting value is measured incorrectly because some of the ion pairs recombine and are not collected. Errors due to the recombination of ion pairs may be especially severe during the measurement of exposure rates from a pulsed beam of x rays.[29]

The range of electrons liberated in air increases rapidly with the energy of incident x or γ rays (see Table 6-9). Electrodes would have to be separated by 4 m in a chamber used to measure x rays generated at 1000 kVp. Above 1000 kVp free-air ionization chambers would become very large, and uniform electric fields would be difficult to achieve. Other problems such as a reduction in the efficiency of ion-pair collection are also encountered with chambers designed for high-energy x and γ rays. With some success, chambers containing air at a pressure of several atmospheres have been designed for high-energy photons. Problems associated with the design of free-air ionization chambers for high-energy x and γ rays contribute to the decision to confine the definition of the roentgen to x and γ rays with energy less than about 3 MeV.

A few corrections are usually applied to measurements with a free-air ionization chamber,[30] including the following:

1. Correction for attenuation of the x or γ rays by air between the diaphragm and the collecting volume
2. Correction for the recombination of ion pairs within the chamber
3. Correction for air density and humidity
4. Correction for ionization produced by photons scattered from the primary beam
5. Correction for loss of ionization caused by inadequate separation of the chamber electrodes

With corrections applied, measurements of radiation exposure with a free-air ionization chamber can achieve accuracies to within $\pm$ 0.5%. As mentioned earlier, free-air ionization chambers are used as reference standards for the calibration of x- and γ-ray beams in standards laboratories. Free-air ionization chambers are too fragile and bulky for routine use.

Thimble Chambers

The amount of ionization collected in a small volume of air is independent of the medium surrounding the collecting volume, provided that the medium has an atomic number equal to the effective atomic number of air. Consequently the large distances required for electron equilibrium in a free-air chamber may be replaced by lesser thicknesses of a more dense material, provided that the atomic number is not changed.

The effective atomic number $\overline{Z}$ of a material is the atomic number of a hypothetical element that attenuates photons at the same rate as the material. For photoelectric interactions, the effective atomic number of a mixture of elements is

$$\overline{Z} = \left(a_1 Z_1^{2.94} + a_2 Z_2^{2.94} + \cdots + a_n Z_n^{2.94}\right)^{\frac{1}{2.94}} \qquad \textbf{(6-20)}$$

where, $Z_1, Z_2, \ldots, Z_n$ are the atomic numbers of elements in the mixture and $a_1, a_2, \ldots a_n$ are the fractional contributions of each element to the total number of electrons in the mixture. A reasonable approximation for effective atomic number may be obtained by rounding the 2.94 to 3 in Eq. (6-20).

Example 6-11

Calculate $\overline{Z}$ for air. Air contains 75.5% nitrogen, 23.2% oxygen, and 1.3% argon. Gram-atomic masses are as follows: nitrogen, 14.007; oxygen, 15.999; and argon, 39.948.

Number of electrons contributed by nitrogen to 1 g air

$$= \frac{(1\text{ g})(0.755)(6.02 \times 10^{23}\text{ atoms/gram-atomic mass})(7\text{ electrons/atom})}{14.007\text{ g/gram-atomic mass}}$$

$$= 2.27 \times 10^{23}\text{ electrons}$$

Number of electrons contributed by oxygen to 1 g air

$$= \frac{(1\text{ g})(0.232)(6.02 \times 10^{23}\text{ atoms/gram-atomic mass})(8\text{ electrons/atoms })}{15.999\text{ g/gram-atomic mass}}$$

$$= 0.70 \times 10^{23}\text{ electrons}$$

Number of electrons contributed by argon to 1 g air

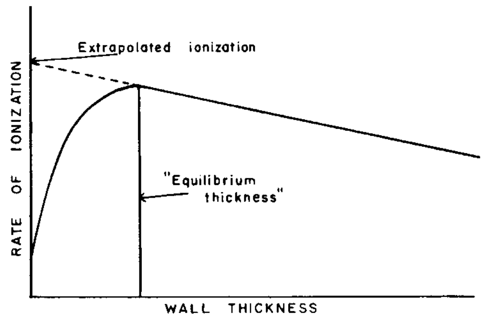
$$= \frac{(1\text{ g})(0.013)(6.02 \times 10^{23}\text{ atoms/gram-atomic mass})(18\text{ electrons/atom})}{(39.948\text{ g/gram-atomic mass})}$$

$$= 0.04 \times 10^{23}\text{ electrons}$$

$$\begin{aligned}
\text{Total electrons in 1 g air} &= (2.27 + 0.70 + 0.04) \times 10^{23} \\
&= 3.01 \times 10^{23}\text{ electrons} \\
a_1\text{ for nitrogen} &= \frac{2.27 \times 10^{23}\text{ electrons}}{3.01 \times 10^{23}\text{ electrons}} = 0.753 \\
a_2\text{ for oxygen} &= \frac{0.70 \times 10^{23}\text{ electrons}}{3.01 \times 10^{23}\text{ electrons}} = 0.233 \\
a_3\text{ for argon} &= \frac{0.04 \times 10^{23}\text{ electrons}}{3.01 \times 10^{23}\text{ electrons}} = 0.013
\end{aligned}$$

$$\begin{aligned}
\overline{Z}_{\text{air}} &= [(0.753)7^{2.94} + (0.233)8^{2.94} + (0.013)18^{2.94}]^{\frac{1}{2.94}} \\
&= 7.64
\end{aligned}$$

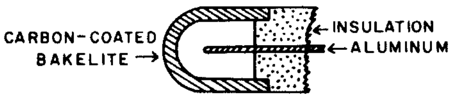
The large distances in air required for electron equilibrium and collection of total ionization in a free-air chamber may be replaced by smaller thicknesses of "air-equivalent material" with an effective atomic number of 7.64. Chambers with air-equivalent walls are known as *thimble chambers*. Most of the ion pairs collected



MARGIN FIGURE 6-9
Ionization in an air-filled cavity exposed to ^{60}Co radiation is expressed as a function of the thickness of the air-equivalent wall surrounding the cavity.

within the air volume are produced by electrons released as photons interact with the air-equivalent walls of the chamber. Shown in Margin Figure 6-9 is the ionization in an air-filled cavity exposed to high-energy x rays, expressed as a function of the thickness of the wall surrounding the cavity. Initially, the number of electrons entering the cavity increases with the thickness of the cavity wall. When the wall thickness equals the range of electrons liberated by incident photons, electrons from outer portions of the wall just reach the cavity, and the ionization inside the cavity is maximum. The thickness of the wall should not be much greater than this equilibrium thickness because the attenuation of incident x and γ rays increases with the wall thickness. The equilibrium thickness for the wall increases with the energy of the incident photons. In Margin Figure 6-9, the slow decrease in ionization as the wall thickness is increased beyond the electron range reflects the attenuation of photons in the wall of the chamber. Extrapolation of this portion of the curve to a wall thickness of zero indicates the ionization that would occur within the cavity if photons were not attenuated by the surrounding wall.

A thimble chamber is illustrated schematically in Margin Figure 6-10. Usually the inside of the chamber wall is coated with carbon, and the central positive electrode is aluminum. The response of the chamber may be varied by changing the size of the collecting volume of air, the thickness of the carbon coating, or the length of the aluminum electrode. It is difficult to construct a thimble chamber with a response at all photon energies identical with that for a free-air chamber. Usually the response of a thimble chamber is compared at several photon energies with the response of a free-air ionization chamber. By using the response of the free-air chamber as a standard, calibration factors may be computed for the thimble chamber at different photon energies.



MARGIN FIGURE 6-10
Diagram of a thimble chamber with an air-equivalent wall.

Condenser Chambers

As ion pairs produced by impinging radiation are collected, the potential difference is reduced between the central electrode and the wall of a thimble chamber. If a chamber with volume v is exposed to X coulomb/kg, the charge Q collected is

$$Q(\text{coulomb}) = X\left(\frac{\text{coulomb}}{\text{kg}}\right)\rho\left(\frac{\text{kg}}{\text{m}^3}\right)v(\text{m}^3)$$

If the density of air is 1.29 kg/m^3, we obtain

$$Q = 1.29Xv$$

For a chamber with capacitance C in farads, the voltage reduction V across the chamber is given by

$$V = \frac{Q}{C} = \frac{1.29Xv}{C}$$

The voltage reduction per unit exposure is

$$\frac{V}{X} = \frac{1.29v}{C}$$

where X is the exposure in coulombs/kg and v is the volume of the chamber in cubic meters. The voltage drop per unit exposure, defined as the *sensitivity* of the chamber, may be reduced by decreasing the chamber volume v or by increasing the chamber capacitance C.

MARGIN FIGURE 6-11
A Keithley model 35020 digital dosimetry system. (Courtesy of Keithley Instruments, Inc.)

If an electrometer with capacitance C_e is used to measure the charge distributed over a condenser chamber with total capacitance C, then the sensitivity of the chamber may be defined as

$$\frac{V}{X} = \frac{1.29v}{C + C_e} \tag{6-21}$$